



Database Management Systems

Relational Algebra - Hands on

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Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

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- **Q1: Retrieve the name and address of all employees who work for the 'Research' department.**
- **Q2: Retrieve the names of employees who have no dependents.**

- **Q1: Retrieve the name and address of all employees who work for the 'Research' department.**

RESEARCH_DEPT $\leftarrow \sigma_{DNAME='Research'}(DEPARTMENT)$

RESEARCH_EMPS $\leftarrow (RESEARCH_DEPT \bowtie_{DNUMBER=DNOEMPLOYEE} EMPLOYEE)$

RESULT $\leftarrow \pi_{FNAME, LNAME, ADDRESS}(RESEARCH_EMPS)$

- **Q2: Retrieve the names of employees who have no dependents.**

ALL_EMPS $\leftarrow \pi_{SSN}(EMPLOYEE)$

EMPS_WITH_DEPS(SSN) $\leftarrow \pi_{ESSN}(DEPENDENT)$

EMPS_WITHOUT_DEPS $\leftarrow (ALL_EMPS - EMPS_WITH_DEPS)$

RESULT $\leftarrow \pi_{LNAME, FNAME}(EMPS_WITHOUT_DEPS * EMPLOYEE)$

As a single expression, these queries become:

- **Q1: Retrieve the name and address of all employees who work for the 'Research' department.**

$$\pi_{\text{Fname, Lname, Address}} (\sigma_{\text{Dname='Research'}} (\text{DEPARTMENT} \bowtie_{\text{Dnumber=Dno}} (\text{EMPLOYEE})))$$

- **Q2: Retrieve the names of employees who have no dependents.**

$$\pi_{\text{Lname, Fname}} ((\pi_{\text{Ssn}} (\text{EMPLOYEE}) - \rho_{\text{Ssn}} (\pi_{\text{Essn}} (\text{DEPENDENT}))) \bowtie \text{EMPLOYEE})$$

Consider a database with the following schema:

Sailors(*sid*: integer, *sname*: string, *rating*: integer, *age*: real)

Boats(*bid*: integer, *bname*: string, *color*: string)

Reserves(*sid*: integer, *bid*: integer, *day*: date)

Write relational algebra expressions for the following queries

<i>sid</i>	<i>sname</i>	<i>rating</i>	<i>age</i>
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5

Figure 4.15 An Instance *S3* of Sailors

<i>sid</i>	<i>bid</i>	<i>day</i>
22	101	10/10/98
22	102	10/10/98
22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

Figure 4.16 An Instance *R2* of Reserves

<i>bid</i>	<i>bname</i>	<i>color</i>
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Figure 4.17 An Instance *B1* of Boats

- 1: Retrieve rows corresponding to expert (rating more than 8) sailors
- 2: Retrieve the rating of each sailor
- 3: Retrieve the age of sailors
- 4: Retrieve name and ratings of highly rated sailor

Find the names of sailors who have reserved boat 103.

$$\pi_{sname}((\sigma_{bid=103}Reserves) \bowtie Sailors)$$

Find the names of sailors who have reserved a red boat.

$$\pi_{sname}((\sigma_{color='red'} Boats) \bowtie Reserves \bowtie Sailors)$$

Find the colors of boats reserved by Lubber.

$$\pi_{color}((\sigma_{sname='Lubber'} Sailors) \bowtie Reserves \bowtie Boats)$$

Find the names of sailors who have reserved at least one boat.

$$\pi_{sname}(Sailors \bowtie Reserves)$$

Find the names of sailors who have reserved a red or a green boat.

$$\rho(Tempboats, (\sigma_{color='red'} Boats) \cup (\sigma_{color='green'} Boats))$$

$$\pi_{sname}(Tempboats \bowtie Reserves \bowtie Sailors)$$

$$\rho(Tempboats, (\sigma_{color='red' \vee color='green'} Boats))$$

$$\pi_{sname}(Tempboats \bowtie Reserves \bowtie Sailors)$$

(Q6) Find the names of sailors who have reserved a red and a green boat.

$$\rho(Tempboats2, (\sigma_{color='red'} Boats) \cap (\sigma_{color='green'} Boats)) \\ \pi_{sname}(Tempboats2 \bowtie Reserves \bowtie Sailors)$$

INCORRECT

$$\rho(Tempred, \pi_{sid}((\sigma_{color='red'} Boats) \bowtie Reserves)) \\ \rho(Tempgreen, \pi_{sid}((\sigma_{color='green'} Boats) \bowtie Reserves)) \\ \pi_{sname}((Tempred \cap Tempgreen) \bowtie Sailors)$$

Find the names of sailors who have reserved at least two boats.

$$\begin{aligned} &\rho(Reservations, \pi_{sid, sname, bid}(Sailors \bowtie Reserves)) \\ &\rho(Reservationpairs(1 \rightarrow sid1, 2 \rightarrow sname1, 3 \rightarrow bid1, 4 \rightarrow sid2, \\ &5 \rightarrow sname2, 6 \rightarrow bid2), Reservations \times Reservations) \\ &\pi_{sname1} \sigma_{(sid1=sid2) \wedge (bid1 \neq bid2)} Reservationpairs \end{aligned}$$

Find the sids of sailors with age over 20 who have not reserved a red boat.

$$\begin{aligned} &\pi_{sid}(\sigma_{age > 20} Sailors) - \\ &\pi_{sid}((\sigma_{color='red'} Boats) \bowtie Reserves \bowtie Sailors) \end{aligned}$$

Find the sids of sailors with age over 20 who have not reserved a red boat.

$$\pi_{sid}(\sigma_{age > 20} Sailors) - \pi_{sid}((\sigma_{color='red'} Boats) \bowtie Reserves \bowtie Sailors)$$

(Q9) Find the names of sailors who have reserved all boats.

$$\rho(Tempsids, (\pi_{sid,bid} Reserves) / (\pi_{bid} Boats)) \\ \pi_{sname}(Tempids \bowtie Sailors)$$

Find the names of sailors who have reserved all boats called Interlake.

$$\rho(Tempsids, (\pi_{sid,bid} Reserves) / (\pi_{bid}(\sigma_{bname='Interlake'} Boats))) \\ \pi_{sname}(Tempids \bowtie Sailors)$$

Consider a database with the following schema:

Person (name, age, gender) name is a key

Frequents (name, pizzeria) (name, pizzeria) is a key

Eats (name, pizza) (name, pizza) is a key

Serves (pizzeria, pizza, price) (pizzeria, pizza) is a key

Write relational algebra expressions for the following queries

- a) Find all pizzerias frequented by at least one person under the age of 18.
- b) Find the names of all females who eat either mushroom or pepperoni pizza (or both).
- c) Find the names of all females who eat both mushroom and pepperoni pizza.
- d) Find all pizzerias that serve at least one pizza that Amy eats for less than \$10.00.
- e) Find all pizzerias that are frequented by only females or only males.
- f) For each person, find all pizzas the person eats that are not served by any pizzeria the person frequents. Return all such person (name) / pizza pairs.
- g) Find the names of all people who frequent only pizzerias serving at least one pizza they eat.
- h) Find the names of all people who frequent every pizzeria serving at least one pizza they eat.
- i) Find the pizzeria serving the cheapest pepperoni pizza. In the case of ties, return all of the cheapest-pepperoni pizzerias.

In general there are many correct expressions for each query.
Sample solutions

a. $\pi_{\text{pizzeria}} \left(\sigma_{\text{age} < 18} (\text{Person}) \bowtie \text{Frequents} \right)$

b. $\pi_{\text{name}} \left(\sigma_{\text{gender} = \text{'female'} \wedge (\text{pizza} = \text{'mushroom'} \vee \text{pizza} = \text{'pepperoni'})} (\text{Person} \bowtie \text{Eats}) \right)$

c. $\pi_{\text{name}} \left(\sigma_{\text{gender} = \text{'female'} \wedge \text{pizza} = \text{'mushroom'}} (\text{Person} \bowtie \text{Eats}) \right) \cap$
 $\pi_{\text{name}} \left(\sigma_{\text{gender} = \text{'female'} \wedge \text{pizza} = \text{'pepperoni'}} (\text{Person} \bowtie \text{Eats}) \right)$

d. $\pi_{\text{pizzeria}} \left(\sigma_{\text{name}='Amy'}(\text{Eats}) \bowtie \sigma_{\text{price} < 10}(\text{Serves}) \right)$

e. $\left(\begin{array}{l} \pi_{\text{pizzeria}} \left(\sigma_{\text{gender}='female'}(\text{Person}) \bowtie \text{Frequents} \right) - \\ \pi_{\text{pizzeria}} \left(\sigma_{\text{gender}='male'}(\text{Person}) \bowtie \text{Frequents} \right) \end{array} \right) \cup$
 $\left(\begin{array}{l} \pi_{\text{pizzeria}} \left(\sigma_{\text{gender}='male'}(\text{Person}) \bowtie \text{Frequents} \right) - \\ \pi_{\text{pizzeria}} \left(\sigma_{\text{gender}='female'}(\text{Person}) \bowtie \text{Frequents} \right) \end{array} \right)$

f. $\text{Eats} - \pi_{\text{name, pizza}}(\text{Frequents} \bowtie \text{Serves})$

g. $\pi_{\text{name}}(\text{Person}) - \pi_{\text{name}}(\text{Frequents} - \pi_{\text{name, pizzeria}}(\text{Eats} \bowtie \text{Serves}))$

h. $\pi_{\text{name}}(\text{Person}) - \pi_{\text{name}}(\pi_{\text{name, pizzeria}}(\text{Eats} \bowtie \text{Serves}) - \text{Frequents})$

i. $\pi_{\text{pizzeria}}(\sigma_{\text{pizza}='pepperoni'} \text{Serves}) -$
$$\pi_{\text{pizzeria}} \left(\sigma_{\text{price} > \text{price2}} \left(\begin{array}{c} \pi_{\text{pizzeria, price}}(\sigma_{\text{pizza}='pepperoni'} \text{Serves}) \\ \times \\ \rho_{\text{pizzeria2, price2}}(\pi_{\text{pizzeria, price}}(\sigma_{\text{pizza}='pepperoni'} \text{Serves})) \end{array} \right) \right)$$

i) alternate solution

$$A \leftarrow \sigma_{\text{pizza}=\text{pepperoni}} (\text{Serves})$$
$$B \leftarrow \rho_{(\text{minprice})}^{\gamma(\text{min}(\text{price}))} (A)$$
$$\pi_{\text{pizzeria}} (\sigma_{\text{price}=\text{minprice}} (A \bowtie B))$$

Relational Algebra

- Unary Relational Operations
- Relational Algebra Operations From Set Theory
- Binary Relational Operations
- Additional Relational Operations
- Examples of Queries in Relational Algebra

1. The relational algebra expression for finding the names of instructors in the Physics department is:

- a) Instructor - Physics
- b) $\sigma \text{ Name } (\pi \text{ Dept_Name} = \text{"Physics"} (\text{Instructor}))$
- c) $\pi \text{ Name } (\text{Instructor} \bowtie \text{Physics})$
- d) $\pi \text{ Name } (\sigma \text{ Dept_Name} = \text{"Physics"} (\text{Instructor}))$

2. The expression $\rho \text{ City} \rightarrow \text{Location} (\text{Customers})$ is used to:

- a) Rename the attribute "City" to "Location" in the "Customers" relation
- b) Select only the "City" column from "Customers"
- c) Find the intersection of "City" and "Location" columns
- d) Perform a join operation on "Customers"

3. The result of the operation π Name, Department (σ JoiningYear < 2020 (Employees)) will return:

- a) EmployeeIDs of employees who joined before 2020
- b) Names and departments of employees who joined before 2020
- c) Employee salaries who joined before 2020
- d) All attributes of employees who joined before 2020

4. The SIDs of students who are not enrolled in any course is given by:

- a) π SID(Student) - π SID(Enrolled)
- b) σ SID NOT IN (π SID(Enrolled)) (Student)
- c) π SID(Student $\bowtie$ Enrolled)
- d) π SID(Enrolled) - π SID(Student)

Answers

1. D
2. A
3. B
4. A



THANK YOU

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